

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

Department of Computer Science and Engineering

CSE-2200: Software Development IV

COMPREHENSIVE PROJECT REPORT

SkillBridge

A Reciprocal Peer-to-Peer Academic Skill Exchange Platform for University Students

GROUP NUMBER: 01

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1. Problem Statement

SkillBridge was conceived to address the persistent challenges faced by individuals seeking academic peer tutoring and collaborative learning within higher education institutions like Ahsanullah University of Science and Technology (AUST). The platform targets three primary user groups affected by an inefficient, fragmented academic assistance ecosystem.

1.1 Affected Stakeholders

- **Peer Learners** – Students facing challenging coursework (e.g., Algorithms, Web Dev, OOP) who face time-consuming searches and a lack of accessible 1-on-1 mentoring.
- **Peer Mentors** – Proficient students willing to mentor peers who lack structured avenues to identify compatible study partners without sacrificing their own study goals.
- **Faculty & Instructors** – Educators observing learning bottlenecks across cohorts who have limited office hours to provide customized tutoring to every struggling student.

1.2 Current Challenges

- Time-consuming searches requiring manual coordination across informal friend groups.
- Lack of centralized platform with complete, filterable syllabus topics and skill tags.
- Financial barriers created by commercial tutoring platforms charging hourly rates.
- Asymmetric skill exchanges causing mentor burnout due to lack of reciprocal support.
- Communication and pacing barriers between students across different semesters.

1.3 Insufficiency of Existing Solutions

Existing approaches fall short in key areas:

Channel	Limitation
Word-of-mouth	Limited reach, highly informal, no verification of subject competence.
Social media groups	Poor search and filter capabilities; listings quickly buried in feeds.
Commercial tutoring	Added fees, focus on profit, misaligned with specific university syllabi.
General online courses	Impersonal, pre-recorded content lacking direct 1-to-1 peer interaction.

2. System Design

The system follows a modern three-tier client-server architecture built on the MERN stack (MongoDB, Express.js, React.js, Node.js). Its decoupled design allows the frontend to operate independently while communicating with backend services via RESTful APIs and encrypted session credentials.

2.1 Frontend – Client Tier

- Single Page Application (SPA) built with React 19, initialized via Vite for fast development and optimized production builds.
- React Router enables seamless client-side navigation without full-page reloads.
- Protected components (<ProtectedRoute>) isolate authenticated views from public onboarding pages.

2.2 Backend – Application Tier

- RESTful API built with Node.js and Express 5 handles all core business logic.
- Non-blocking asynchronous I/O execution guarantees high concurrency and responsive throughput.

2.3 Database – Data Tier

- MongoDB document database provides a flexible schema suitable for student profiles and skill taxonomies.
- Mongoose ODM defines and manages relationships between Users, Tags, Matches, Messages, and Notifications.

2.4 Authentication & Security

- Stateless JWT-based authentication ensures scalable, sessionless security (15-minute access tokens).
- HTTP-only SameSite=Lax cookies store refresh tokens, mitigating Cross-Site Scripting (XSS) attack surfaces.
- Passwords are securely hashed using the bcryptjs algorithm with adaptive salt rounds.
- Double-submit CSRF protection validates all state-modifying requests (POST, PUT, DELETE).
- Payload limiters (10kb) and strict CORS origin whitelisting defend against DoS and unauthorized cross-origin requests.
- Credential masking middleware redacts tokens and sensitive data from application logs.

2.5 Matchmaking & Real-Time Communication Layer

To support equitable peer collaboration, SkillBridge integrates dedicated services for reciprocal pairing and messaging:

- **2-Cycle Reciprocal Matchmaker** evaluates bipartite complementarity to ensure mutual academic value before pair generation.
- **Match-Scoped Messaging Engine** guarantees private 1-to-1 communication channels restricted to approved study matches.
- **Deep-Link URL Resolvers** (/chat?peerId=...&matchId=...) ensure seamless state transitions across views.

3. Implementation Details

3.1 Technology Stack

Layer	Technology	Purpose
Frontend	React 19 + Vite 8	Component-based UI, fast HMR, optimized production builds.
Styling	Tailwind CSS v4	Utility-first responsive design, modern badge UI, zero runtime overhead.
Routing	React Router v7	Client-side navigation and protected authentication routes.
Backend	Node.js + Express 5	RESTful API, asynchronous business logic, modular middleware.
Database	MongoDB + Mongoose 9	Flexible document storage, typed schemas, relationship population.
Auth	JWT + bcryptjs	Stateless authentication (15m access / 7d refresh), secure password hashing.

Layer	Technology	Purpose
Security	Double-Submit CSRF + Cookies	Defense-in-depth protection against XSS, CSRF, and payload inflation.
Sustainability	@tgwf/co2	Carbon emission estimation per HTTP session (SWD model).
Client Eco-Track	react-carbon-footprint	Real-time client-side network byte and CO ₂ emission monitoring.
Testing	Node.js Assert Harness	Automated unit tests, token validation, and matchmaking algorithm tests.

3.2 Core Feature Implementation

User Authentication & Dual-Token Security

- The `/api/v1/auth` endpoints verify credentials, evaluate bcrypt hashes, and issue short-lived access JWTs paired with long-lived HTTP-only refresh cookies.
- All protected API requests validate the access token on each call; automatic silent refresh preserves sessions seamlessly.

Reciprocal 2-Cycle Matchmaking Engine

- Unlike conventional one-way tutor discovery systems, SkillBridge strictly enforces **2-cycle academic reciprocity**:

```
Overlap_Teach = UserA.strongTags ∩ UserB.weakTags ≠ ∅
Overlap_Learn = UserB.strongTags ∩ UserA.weakTags ≠ ∅
```

- Eligible candidates are ranked by a multi-factor compatibility scoring formula:

```
Score = (|Overlap_Teach| × 3) + (|Overlap_Learn| × 3) - (|ΔSemester| × 1) + (SameDeptBonus × 2)
```

- Exclusion filters automatically omit self-matches, existing pending requests, and already accepted connections.

Academic Taxonomy & Student Profile Management

- Students customize profiles with department, semester, bio, and weekly availability schedules.
- Taxonomy tags span core domains: *Languages*, *Web Dev*, *Mobile Dev*, *Core CS*, and *AI/Data Science*. Dynamic tag resolution allows instant creation of new syllabus-specific tags.

Real-Time 1-to-1 Peer Chat

- Enforces match-scoped message storage referencing `matchId`, sender, recipient, timestamp, and read status.
- Features conversation search filtering, unread counter badges, and deep-link query parameter parsing (`/chat?peerId=...`).

Carbon Tracking Middleware

- An Express middleware intercepts responses and measures total bytes transferred per session (body, headers, queries).
- Emissions are estimated using the `@tgwf/co2` model, converting byte counts to CO₂ grams.
- Example session measurement: 17,722 bytes transferred, resulting in approximately 0.0026 to 0.0068 grams of CO₂ (~3–7 mg per session).

Security Middleware Stack

- A custom authentication middleware enforces token verification on all protected endpoints, preventing unauthorized resource access.
- Double-submit CSRF verification rejects state mutations lacking valid matching tokens, complemented by a strict CORS policy whitelisting only verified frontend origins.
- 35 automated tests validate system integrity (25 unit/security tests and 10 matchmaking algorithm tests).

4. Societal Impact Analysis

4.1 Impact on Peer Learners

- **Saves Significant Study Time** by eliminating inefficient, unorganized manual searches across informal chat groups.
- **Provides Comprehensive Syllabus Matching** including exact course topics, programming languages, and peer availability.
- **Eliminates Exploitative Tutoring Fees** by establishing an equitable, barter-based reciprocal skill exchange.
- **Direct Peer Communication** reduces anxiety, allowing students to ask foundational questions in a supportive environment.

4.2 Impact on Peer Mentors

- **Reinforces Knowledge via the Protégé Effect:** Mentoring peers deepens the mentor's conceptual mastery and debugging intuition.
- **Guarantees Reciprocal Assistance:** Ensures high-performing students receive help in their own growth areas, preventing burnout.
- **Cultivates Leadership Skills:** Builds interpersonal communication, code review abilities, and professional mentorship habits.

4.3 Broader Community Benefits

- **Promotes a Collaborative Campus Culture:** Fosters mutual support rather than zero-sum academic competition at AUST.
- **Bridges Departmental & Semester Silos:** Connects junior and senior students across computing disciplines.
- **Improves Course Retention:** Provides an accessible safety net for challenging gateway courses like Algorithms and OOP.

5. Carbon Footprint Analysis

SkillBridge incorporates Sustainable Web Design principles directly into its server architecture. This reflects a commitment to environmentally responsible digital services.

5.1 Methodology

- An Express.js middleware intercepts all outgoing HTTP responses and incoming payloads, measuring bytes transferred.
- The `@tgwf/co2` library converts byte transfer data into estimated CO₂ emissions using the Sustainable Web Design (SWD) model.
- The frontend integrates `react-carbon-footprint` to monitor client network transfer and render a live eco-indicator.
- Carbon data is logged per session and attached as response headers (X-Bytes-Transferred, X-Estimated-CO2-Grams).

5.2 Measurement Example

Metric / Scenario	Data Transferred	Estimated CO ₂
Bytes Transferred (session benchmark)	17,722 bytes	0.0026–0.0068 g (~3–7 mg)
Discovery Dashboard Feed & Tags	14,680 bytes	0.0022 grams (~2.2 mg)
Active Peer Chat (15 Real-time Msgs)	18,450 bytes	0.0027 grams (~2.7 mg)
Production SPA Bundle (Vite Gzipped)	113,740 bytes	0.0168 grams (~16.8 mg)
Typical End-to-End User Session	24,200 bytes	0.0036 grams (~3.6 mg)
Sustainability Model	@tgwf/co2 & react-carbon-footprint (SWD Model)	

5.3 Environmental Significance

- **Demonstrates Digital Carbon Awareness:** Demonstrates that web applications can be built with explicit awareness of their carbon footprint.
- **Encourages Efficient Data Transfer:** Encourages efficient data transfer practices, reducing unnecessary payload sizes.
- **Vite 8 Build Optimization:** Code-splitting and Rollup tree-shaking compress production assets by over 73%, directly reducing global transmission emissions.
- **Positions SkillBridge as an Eco-Conscious Model:** Serves as a model for eco-conscious software development within the local tech ecosystem.

6. Conclusion

SkillBridge represents a meaningful step toward solving a well-documented, real-world problem in university academic environments. By consolidating peer discovery, mutual skill verification, direct messaging, and exchange tracking into a single cohesive platform, it delivers clear and measurable value to peer learners, student mentors, and the broader university community.

Built on a robust MERN stack with JWT-secured dual-token authentication, real-time messaging, and cloud-based data management, the technical implementation reflects industry best practices in modern web development. The mathematical formulation of the 2-cycle reciprocal matchmaking engine ensures that academic cooperation is fundamentally equitable, structured, and mutually beneficial.

The integration of carbon footprint tracking via an Express middleware and the react-carbon-footprint client library goes beyond functionality, demonstrating a commitment to environmentally responsible engineering — a distinction that sets SkillBridge apart from comparable platforms.

The platform does not merely replicate existing communication tools; it addresses their specific shortcomings by providing categorized syllabus filtering, verified bilateral matching, private communication channels, and a transparent learning exchange that empowers students to learn together sustainably.

Course Evaluation & Verification Summary

Course	CSE-2200: Software Development IV / Web Development
Institution	Department of Computer Science and Engineering, Ahsanullah University of Science and Technology
Deliverables	Hard Copy & Soft Copy Report (Problem Statement, System Design, Implementation, Societal Impact, Carbon Footprint)
Automated Test Suite	35 Passing Tests (25 Unit & Security Tests, 10 Matchmaking Algorithm Tests)
Sustainability Model	Sustainable Web Design (SWD) Model via @tgwf/co2 and react-carbon-footprint